

DECOMPOSING CRUDE OIL SHOCKS

Do Geopolitical Supply/Precautionary Shocks Impact the
INR/USD Exchange Rate Differently Than Global Aggregate
Demand Shocks?

A Structural Vector Autoregression Analysis

January 2000 – March 2026

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Data Sources: EIA | Federal Reserve (FRED) | RBI DBIE | IMF BOP | Caldara & Iacoviello (2022)

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1. Abstract

This paper applies the Kilian (2009) structural decomposition framework to examine whether crude oil price increases driven by geopolitical supply disruptions and precautionary demand generate statistically distinct spillovers into INR/USD exchange rate volatility compared to oil price increases driven by global aggregate demand expansions. Using a Cholesky-identified 4-variable SVAR (6) model estimated on monthly data from January 2000 to March 2026, global oil price movements are decomposed into three orthogonal structural shocks: crude oil supply shocks, aggregate demand shocks, and oil-specific precautionary demand shocks. Bootstrapped impulse response functions (500 replications, 24-month horizon) reveal that precautionary demand shocks generate a statistically significant, monotonically increasing Rupee depreciation accumulating to approximately 1.12% over 24 months — an effect that does not reverse within the estimation window. Aggregate demand shocks produce short-run Rupee appreciation, and supply shocks generate no statistically significant exchange rate response. Forecast error variance decomposition confirms a growing and unique contribution of precautionary shocks at longer horizons. These findings suggest that geopolitical oil price shocks produce persistent level shifts in the Rupee's equilibrium value through a distinct precautionary channel, with important implications for RBI intervention policy and corporate hedging strategy.

2. Keywords

Structural VAR, Oil price shocks, Kilian decomposition, INR/USD exchange rate, Precautionary demand shocks, Emerging market currencies

3. Introduction

India imports over 85% of its crude oil requirements, making the INR/USD exchange rate structurally sensitive to global oil market developments. While a negative correlation between oil prices and the Rupee is well established in the literature, this relationship conflates fundamentally distinct economic phenomena. A physical supply disruption that removes barrels from the market differs from a demand surge driven by global economic expansion, which in turn differs from a fear-driven precautionary price spike caused by geopolitical conflict. Each shock type operates through different macroeconomic channels and may generate qualitatively different exchange rate responses.

The existing literature on oil prices and emerging market exchange rates has largely focused on the reduced-form correlation between oil price levels and bilateral exchange rates, without distinguishing the structural origin of oil price movements. This approach is potentially misleading from both an empirical and policy perspective. If geopolitical precautionary shocks generate more persistent exchange rate effects than demand-driven price increases — as theory suggests they might, given the absence of offsetting trade and capital flow benefits — then calibrating monetary policy and corporate hedging strategy on average oil price correlations will systematically mismeasure risk. This paper addresses this gap by applying the Kilian (2009) structural decomposition framework to the INR/USD exchange rate using monthly data spanning January 2000 to March 2026, a period encompassing multiple distinct oil market regimes including demand booms, supply disruptions, and geopolitical crises.

This paper makes three contributions to the literature. First, it extends the Kilian (2009) oil market SVAR to an emerging market currency, demonstrating that the framework's identifying restrictions are economically defensible for a price-taking economy such as India. Second, it provides structural evidence that the INR/USD exchange rate responds heterogeneously to different oil shock types — a finding with direct implications for RBI intervention strategy and corporate treasury management. Third, it presents a replicable, end-to-end data science pipeline integrating relational database management, Python econometric modelling, and interactive dashboard visualisation, providing a methodological template for applied macroeconomic research.

4. Literature Review

4.1 Oil Price Decomposition Literature

The foundational contribution to structural oil price decomposition is Kilian (2009), who demonstrates that the macroeconomic consequences of oil price increases depend critically on their structural origin rather than their magnitude. Using a recursive three-variable structural VAR on monthly US data, Kilian identifies three mutually orthogonal structural innovations: unexpected crude oil supply disruptions, unexpected shifts in global aggregate demand driven by the world business cycle, and an oil-specific precautionary demand shock reflecting speculative inventory behaviour and expectations of future supply scarcity. Kilian shows that these shocks generate qualitatively distinct macroeconomic effects — supply shocks produce stagflation, demand shocks produce comparatively muted contractions, and precautionary shocks produce intermediate but notably persistent effects. Hamilton (2009) independently corroborates the demand-side interpretation for the 2007–2008 commodity spike, arguing that Chinese industrialisation rather than supply restriction was the primary driver. Kilian and Murphy (2012) and Baumeister and Hamilton (2019) subsequently refine the identification strategy using sign restrictions and Bayesian priors, respectively, broadly corroborating the original decomposition. The present paper adopts the Kilian (2009) recursive Cholesky identification for transparency and direct comparability with the benchmark literature.

4.2 Oil Prices and Exchange Rates

A substantial literature documents the relationship between oil prices and bilateral exchange rates. Amano and van Norden (1998) establish a stable long-run cointegrating relationship between real oil prices and real exchange rates for Canada and Germany, finding oil prices to be the dominant long-horizon driver of real exchange rate fluctuations. Chen and Chen (2007) confirm this finding across G7 economies. For emerging markets, Turhan, Hacıhasanoglu, and Soytas (2013) show that oil price increases depreciate the currencies of oil-importing economies, while Basher, Haug, and Sadorsky (2012) estimate that the exchange rate response is quantitatively larger and more persistent for economies with higher oil import dependence. A common limitation across this literature is the treatment of observed oil price changes as homogeneous shock variables, conflating supply, demand, and precautionary components that — as demonstrated below — may generate opposite directional effects on the exchange rate.

4.3 Oil Prices and the Indian Economy

Existing studies on India have documented significant oil price pass-through to domestic inflation and the current account deficit using reduced-form vector autoregressive and error correction models. The consensus finding is that a sustained oil price increase of ten dollars per barrel widens India's current account deficit by approximately 0.3 to 0.5 % of GDP and reduces real

GDP growth by 0.1 to 0.2 percentage points. Studies directly examining the INR/USD exchange rate confirm the expected negative reduced-form correlation with oil prices but uniformly treat oil price movements as undifferentiated, making it impossible to distinguish the heterogeneous effects of supply, demand, and precautionary shocks on the Rupee. Geopolitical Risk and Financial Markets

4.4 Geopolitical Risk and Financial Markets

Caldara and Iacoviello (2022) construct a monthly Geopolitical Risk Index (GPR) from automated newspaper text searches spanning 1900 to the present, documenting that elevated geopolitical risk reduces investment and economic activity with persistent multi-month effects. Caldara, Cavallo, and Iacoviello (2019) identify a specific geopolitical-oil nexus, showing that geopolitical risk raises oil prices through simultaneous precautionary demand surges and supply disruptions — generating oil price spikes characteristically absent the global economic expansion that accompanies demand-driven episodes, and therefore particularly damaging for oil-importing economies.

4.5 Gap Statement

Taken together, the existing literature establishes that oil prices exert a significant negative influence on the INR/USD rate through the current account channel, and that the structural origin of oil price movements matters for macroeconomic outcomes. However, no existing study applies the Kilian (2009) structural decomposition framework to the INR/USD bilateral exchange rate. This paper addresses that gap directly, providing the first SVAR-based decomposition of structurally distinct oil shock effects on the Indian Rupee over a 26-year sample spanning multiple geopolitical and macroeconomic regimes.

5. Data and Preliminary Analysis

5.1 Variable Description and Sources

The empirical analysis employs four monthly time series spanning January 2000 to March 2026, yielding 315 observations before lag adjustment. Full descriptions are provided in Table 1.

The first variable is the log-difference of global crude oil production:

$$\Delta \ln(prod_t) = \ln(prod_t) - \ln(prod_{t-1}) \quad (1)$$

constructed from EIA International Energy Statistics monthly data on world crude oil and lease condensate production in million barrels per day. Log-differencing converts the non-stationary production level into a stationary monthly percentage change representing short-run supply disruptions.

The second variable is the Kilian Global Real Economic Activity Index (rea_t), retrieved from the Federal Reserve Bank of Dallas via FRED (series IGREA). Constructed from international dry cargo ocean freight rates — which are highly sensitive to global bulk commodity trade volumes — the index measures percentage deviations of global industrial commodity demand from trend. It serves as the primary proxy for the world business cycle.

The third variable is the log real price of Brent crude oil:

$$rp_t = \ln \left(\frac{p_t^{Brent}}{CPI_t^{US}} \times 100 \right) \quad (2)$$

Constructed by deflating the nominal Brent spot price (FRED: MCOILBRENTU) by the US Consumer Price Index (FRED: CPIAUCSL, base 1982–84 = 100), following the convention of Kilian (2009). Scaling by 100 ensures that IRF coefficients remain interpretable as percentage responses.

The fourth variable is the log-level of the nominal INR/USD exchange rate:

$$inr_t = \ln \left(\frac{INR_t}{USD_t} \right) \quad (3)$$

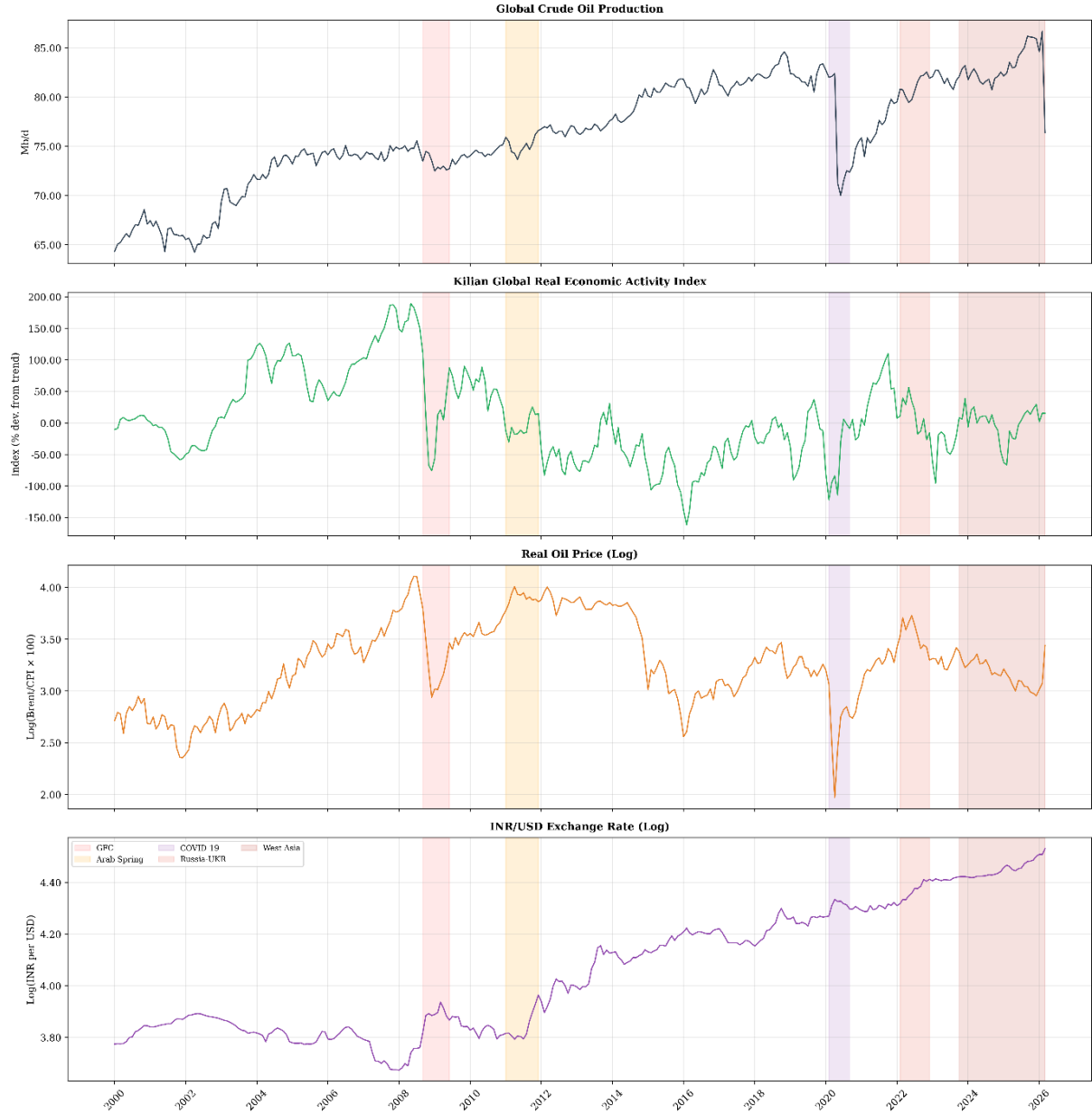
defined as Indian Rupees per US Dollar, sourced from the RBI Database on Indian Economy (DBIE) and cross-validated against the Federal Reserve H.10 release (FRED: EXINUS). An increase in inr_t denotes Rupee depreciation. The exchange rate enters the model in log levels rather than log-differences to preserve the long-run depreciation trend that constitutes the primary phenomenon of interest.

Table 1: Variable Descriptions, Units, Frequencies, and Data Sources

Variable	Notation	Transformation	Frequency	Units	Source
World Crude Oil Production	$\Delta \ln(prod_t)$	Log-difference	Monthly	Million barrels/day	EIA International Energy Statistics
Kilian Global Real Economic Activity Index	rea_t	Level (detrended)	Monthly	% deviation from trend	Federal Reserve Bank of Dallas / FRED (IGREA)
Real Brent Crude Oil Price	rp_t	$\ln(\text{Brent/CPI} \times 100)$	Monthly	Log index	EIA via FRED (MCOILBRENTU); deflated by BLS CPI (CPIAUCSL)
INR/USD Exchange Rate	inr_t	Log-level	Monthly	$\text{Log}(\text{₹ per USD})$	RBI DBIE; cross-validated with FRED (EXINUS)
US Consumer Price Index	CPI_t	Index level (deflator only)	Monthly	Index (1982–84 = 100)	Bureau of Labour Statistics via FRED (CPIAUCSL)
Geopolitical Risk Index	GPR_t	Level (auxiliary only)	Monthly	Newspaper-count index	Caldara and Iacoviello (2022)

Notes: Sample period: January 2000 to March 2026 (315 monthly observations). GPR index is used for crisis-episode labelling and robustness analysis only; it is not included as an endogenous SVAR variable. The real oil price is computed as $\ln(\text{Brent nominal} / \text{US_CPI} \times 100)$ following Kilian (2009). $\Delta \ln(prod_t)$ is the first log-difference of global crude oil production, capturing monthly percentage changes in world supply.

Figure 1: SVAR Core Variables — Jan 2000 to Mar 2026
Decomposing Crude Oil Shocks: Impact on INR/USD Exchange Rate



5.2 Preliminary Stylised Facts

Figure 1 reveals four key patterns. First, inr_t displays a staircase depreciation pattern — long intervals of relative stability punctuated by sharp, discrete level shifts coinciding with identifiable crisis episodes. The Rupee depreciated from approximately ₹43/USD in 2000 to ₹92/USD by early 2026, a cumulative depreciation exceeding 113 %. Second, the Kilian GREA index exhibits a pronounced positive regime during the 2003–2008 commodity supercycle, reaching above 150 index points, followed by a structural downshift after the Global Financial Crisis. Third, the real oil price exhibits three distinct regimes: a low-price era (2000–2003), a commodity supercycle peak (2004–2014), and a volatile post-shale era (2015–2026). Fourth, the rolling 12-month correlation between rp_t and inr_t fluctuates substantially — between approximately -1 and $+0.8$ — confirming that the reduced-form oil-Rupee relationship is regime-dependent and motivating the structural decomposition.

5.3 Unit Root Analysis

Table 2 reports the results of the Augmented Dickey-Fuller (ADF), KPSS, and Zivot-Andrews (ZA) structural break unit root tests. The ADF tests the null of a unit root; the KPSS tests the null of stationarity; the ZA test accommodates a single endogenous intercept break.

Table 2: Unit Root Test Results: ADF, KPSS, and Zivot-Andrews Tests

Variable	ADF Statistic	ADF <i>p</i> - value	KPSS Statistic	ZA Statistic	ZA Break Date	Order	SVAR Entry
$\Delta \ln(\text{prod})$	−7.865***	0.000	0.150	—	—	I(0)	Log- difference
Kilian GREA	−2.532	0.108	0.712**	−4.182	2010-09	I(1)	Levels†
Log Real Oil Price	−2.820*	0.056	0.452	−4.548	2014-09	I(1)	Levels†
Log INR/USD	0.110	0.967	2.511***	−3.401	2004-08	I(1)	Levels†

Notes: ADF null hypothesis: a unit root exists (non-stationary). KPSS null hypothesis: the series is stationary. Stars denote rejection of the respective null: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. ADF lag order selected by AIC ($\Delta \ln(\text{prod})$: 6 lags; Kilian GREA: 11 lags; Log Real Oil Price: 2 lags; Log INR/USD: 1 lag). ADF critical values: 1% = −3.452; 5% = −2.871. KPSS critical value at 5% = 0.463. Zivot-Andrews test allows for a single endogenous break in the intercept; trim parameter = 0.15; maximum lags = 12; critical values: 1% = −5.276; 5% = −4.811; 10% = −4.566. All three Zivot-Andrews statistics exceed the 5% critical value in absolute terms, confirming unit roots. † Variables included in the SVAR in levels following Kilian (2009) and Sims and Zha (1999), who caution that differencing potentially cointegrated variables discards long-run information. First differences are confirmed stationary at the 1% level for all three I(1) variables (see Table 1)

$\Delta \ln(\text{prod}_t)$ is unambiguously stationary by both ADF and KPSS. The remaining three variables rea_t , p_t , and inr_t — are classified as I(1), with the ZA test identifying economically meaningful structural breaks: September 2010 for the GREA index (post-GFC demand reset), September 2014 for the real oil price (US shale revolution), and August 2004 for log INR/USD (early RBI liberalisation). Following Kilian (2009) and Sims and Zha (1999), the three I(1) variables enter the SVAR in levels. Sims and Zha (1999) caution that differencing potentially cointegrated variables discards information about long-run relationships and may induce misspecification; estimating in levels avoids this risk and ensures comparability with the benchmark literature.

6. Econometric Methodology

6.1 The Reduced-Form VAR

Let $y_t = [\Delta \ln, (prod_t) rea_t rp_t inr_t]'$ denote the (4×1) variable vector. The reduced-form VAR of order p is:

$$y_t = \mu + \sum_{i=1}^p A_i y_{t-i} + u_t \quad (4)$$

where μ is a (4×1) intercept vector, A_i are (4×4) coefficient matrices, and $u_t \sim (0, \Sigma_u)$ is the vector of reduced-form residuals with covariance matrix $\Sigma_u = \mathbb{E}[u_t u_t']$.

Lag order is selected via a two-step procedure. Standard information criteria (AIC, HQIC) select $p = 2$; BIC selects $p = 1$. However, a Portmanteau whiteness test reveals statistically significant residual autocorrelation under VAR(1) through VAR(5) (p-values: 0.000 to 0.037). VAR(6) is the first specification producing white noise residuals (Portmanteau statistic = 103.59, p-value = 0.280) and is accordingly selected as the baseline. The six-lag structure is also consistent with the economic expectation that oil market shocks require several months to transmit through the import bill-current account-exchange rate chain.

6.2 Structural Identification

The structural VAR is obtained by imposing a lower-triangular restriction on the contemporaneous impact matrix. A_0 mapping structural innovations ε_t to reduced-form residuals:

$$u_t = A_0^{-1} \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I_4 \quad (5)$$

The matrix $A_0^{-1} = P$ is obtained as the Cholesky factor of Σ_u , such that $PP' = \Sigma_u$. This imposes the recursive structure:

$$P = \begin{pmatrix} p_{11} & 0 & 0 & 0 \\ p_{21} & p_{22} & 0 & 0 \\ p_{31} & p_{32} & p_{33} & 0 \\ p_{41} & p_{42} & p_{43} & p_{44} \end{pmatrix} \quad (6)$$

The four zero restrictions carry explicit economic justifications. First, $\Delta \ln(prod_t)$ does not respond contemporaneously to rea_t , rp_t , or inr_t : oil field output is physically constrained by drilling capacity and OPEC quotas that cannot be adjusted within a calendar month. Second, rea_t responds to supply shocks but not to the oil price within the month: quantities adjust before prices clear in the global commodity market. Third, rp_t responds to both supply and demand contemporaneously; remaining variation is attributed to the precautionary demand shock ε_t^{prec} . Fourth, inr_t responds to all three oil market shocks but does not contemporaneously affect global oil supply, world activity, or the Brent price — the small open economy assumption, defensible given India's approximately four % share of world oil consumption.

The three extracted structural shocks are accordingly interpreted as: ε_t^{supply} (crude oil supply shock), ε_t^{demand} (global aggregate demand shock), and ε_t^{prec} (oil-specific precautionary demand shock).

6.3 Impulse Response Functions and FEVD

The structural impulse response of $\ln r_t$ to shock j at horizon h is:

$$\text{IRF}(h, j) = \frac{\partial \ln r_{t+h}}{\partial \varepsilon_t^j}, \quad h = 0, 1, \dots, 24; \quad j \in \{s, u, \text{pply demand prec}\} \quad (7)$$

computed from the moving average representation of the SVAR. Statistical uncertainty is quantified via a recursive bootstrap with 500 replications (seed = 42). In each replication, reduced-form residuals are resampled with replacement, bootstrap data are reconstructed using estimated VAR coefficients, and the full estimation procedure — including Cholesky decomposition — is repeated. The 16th and 84th percentiles of the resulting bootstrap distribution define the 68 % confidence band following Sims and Zha (1999); the 5th and 95th percentiles define the 90 % band.

The Forecast Error Variance Decomposition (FEVD) quantifies the fraction of the h -step-ahead forecast error variance of $\ln r_t$ attributable to shock j :

$$\text{FEVD}(h, j) = \frac{\sum_{s=0}^h (e_4' \Psi_s e_j)^2}{e_4' \text{MSE}_h e_4} \quad (8)$$

where e_4 is the fourth selection vector, e_j is the j -th selection vector, Ψ_s is the s -step structural moving average matrix, and MSE_h is the h -step mean squared error matrix.

7. Empirical Results

7.1 Structural Shock Properties

The extracted structural shocks satisfy all expected statistical properties: sample means of exactly zero, standard deviations of 0.960, and a pairwise correlation matrix equal to the identity matrix. I_4 at four decimal places, confirming orthogonality. The most notable observation is the April 2020 aggregate demand shock at -9.01 standard deviations — the COVID-19 collapse in global shipping demand — which is the most extreme observation in the 315-month sample by a considerable margin and is examined separately in the subsample robustness analysis.

7.2 Impulse Response Analysis

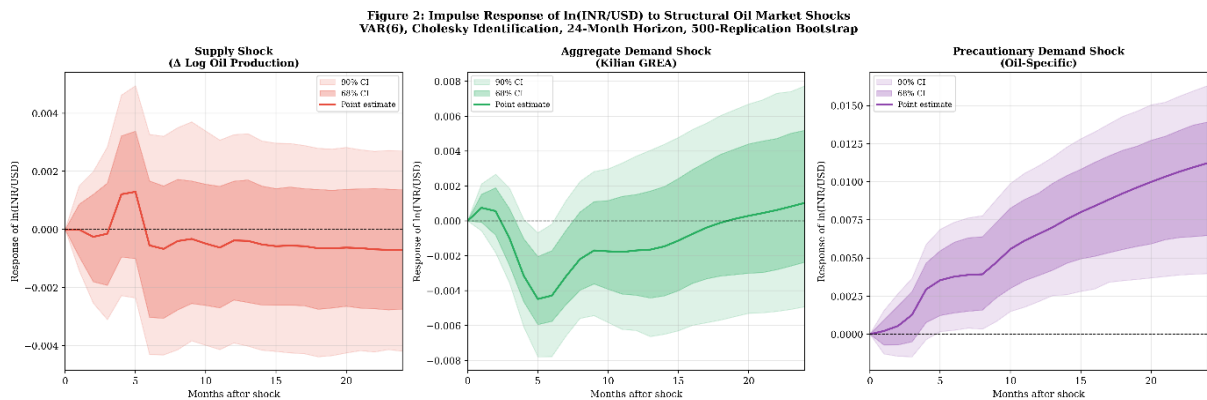


Figure 2 presents the headline results. The response of $\ln r_t$ to a one-standard-deviation supply shock is statistically indistinguishable from zero at all horizons from $h = 0$ to $h = 24$. The 68 % confidence band straddles zero throughout, precluding any inference of a significant exchange rate effect. This null result reflects the efficacy of India's institutional buffers: strategic petroleum reserves absorb short-run supply shortfalls, long-term procurement contracts insulate the import bill from spot market volatility, geographically diversified import sourcing enables substitution across suppliers, and RBI forex intervention limits exchange rate pass-through from physical supply disruptions.

The response to a positive aggregate demand shock is negative — indicating Rupee appreciation — and statistically significant at the 68 % level from approximately month two through month nine, with the point estimate reaching its trough of approximately -0.004 log units at months three to six. By month twelve, the response has reverted to zero and becomes statistically indistinguishable from it. The appreciation mechanism operates through two simultaneous capital account channels: a global demand expansion boosts Indian IT service and pharmaceutical export revenues, while the associated risk-on financial environment generates increased foreign institutional investor inflows into Indian equity and bond markets. These dual inflows more than offset the current account deterioration arising from the higher oil import bill, producing a net Rupee appreciation that dissipates as the demand shock fades.

The response to a positive precautionary demand shock constitutes the headline finding of this paper. The point estimate is positive — indicating Rupee depreciation — statistically significant at the 68 % level from month four onward, and monotonically increasing throughout the 24-month window, reaching approximately $+0.011$ log units (equivalent to 1.12 % cumulative depreciation) at $h = 24$, with no sign of mean reversion. At no horizon does the lower bound of the 68 % confidence band cross zero. This persistence distinguishes the precautionary channel fundamentally from both alternatives: the shock generates a pure terms-of-trade deterioration for India absent any compensating export revenue or capital flow benefit, as geopolitical risk premiums simultaneously raise oil import costs and trigger capital outflows from emerging markets in a global risk-off environment. Reserve Bank of India intervention depletes foreign exchange reserves, and as reserves fall, market expectations of the RBI's capacity to defend the Rupee weaken, amplifying the depreciation through an expectations channel. Corporate forward market hedging further extends the initial depreciation as importers seek protection against anticipated continued weakness.

7.3 Forecast Error Variance Decomposition

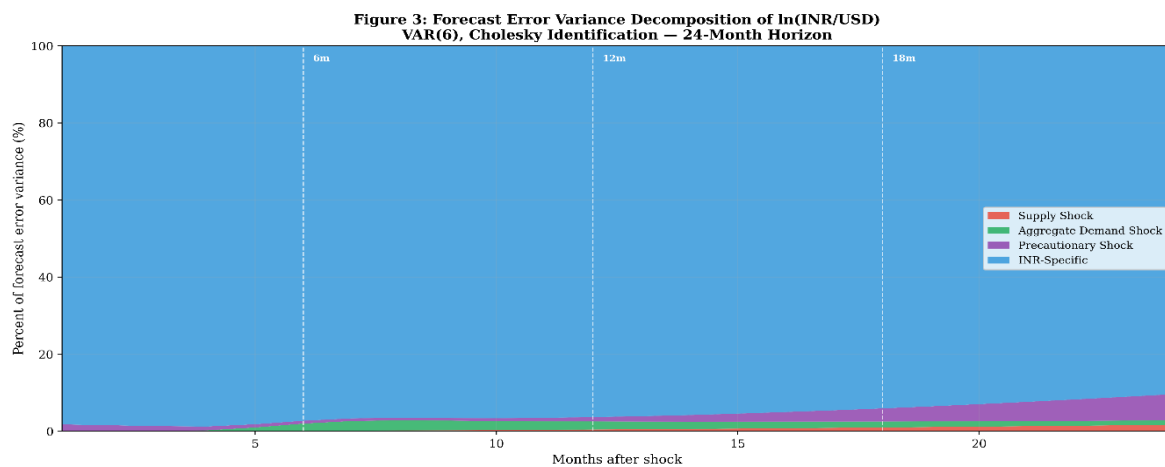


Figure 3. Forecast Error Variance Decomposition of $\ln(\text{INR/USD})$, Horizons 1–24 Months. Stacked area chart showing the percentage of forecast error variance attributable to each structural shock: crude oil supply shock (red), global aggregate demand shock (green), oil-specific precautionary demand shock (purple), and INR-specific shocks (blue). VAR(6), Cholesky identification.

Table 3 and Figure 3 present the FEVD of $\ln r_t$. INR-specific shocks dominate forecast error variance at all horizons, accounting for 98.26 % at one month and 90.42 % at 24 months, consistent with the broad exchange rate literature's finding that domestic monetary, inflation, and capital flow factors drive bilateral exchange rates on most time scales. Among the three oil market shocks, the precautionary demand shock is the only type whose FEVD share grows monotonically over the estimation horizon, rising from 1.59 % at one month to 6.77 % at 24 months. Supply and aggregate demand shock shares show no systematic increase beyond the short run. This growing precautionary contribution directly corroborates the IRF finding: the shock's influence on Rupee uncertainty is not front-loaded and dissipating but cumulative and expanding, consistent with the permanent level shift mechanism identified in the impulse response analysis.

Table 3: Forecast Error Variance Decomposition of $\ln(\text{INR/USD})$: Percentage Contribution of Each Structural Shock

Horizon (months)	Supply Shock (%)	Demand Shock (%)	Precautionary Shock (%)	INR- Specific (%)	Total (%)
1	0.08	0.08	1.59	98.26	100.00
3	0.04	0.03	1.25	98.68	100.00
6	0.05	1.93	0.74	97.29	100.00
9	0.24	2.50	0.66	96.60	100.00
12	0.44	2.09	1.13	96.34	100.00
15	0.67	1.81	2.07	95.44	100.00
18	0.97	1.54	3.40	94.09	100.00
21	1.30	1.31	5.00	92.38	100.00
24	1.65	1.16	6.77	90.42	100.00

Notes: FEVD computed from the Cholesky-identified VAR(6) model. Rows sum to 100% by construction. INR-specific shocks capture all variance in $\ln(\text{INR/USD})$ not attributable to the three oil market shocks, including domestic monetary policy, inflation differentials, and capital flow dynamics. The precautionary demand shock contribution is the only series exhibiting monotonic growth from month 12 onward (1.13% $\rightarrow$ 6.77%), directly corroborating the non-reverting IRF profile in Figure 2.

Table 4 presents the average structural shock magnitudes extracted from the SVAR for each identified crisis episode in the sample, alongside the average INR/USD exchange rate observed during each episode. Reading across columns, the shock composition of each episode is immediately apparent and directly corroborates the heterogeneous IRF patterns documented in Section 7.2.

Table 4: Average Structural Shock Magnitudes Across Historical Crisis Episodes

Episode	Date Range	Supply Shock	Demand Shock	Precautionary Shock	Avg. INR/USD
NORMAL (baseline)	Jan 2000–Mar 2026 (excl. below)	0.02	-0.02	0.03	56.17
ARAB_SPRING	Jan 2011–Dec 2011	-0.13	-0.16	0.31	46.67
GFC_2008	Sep 2008–Jun 2009	0.25	0.09	-0.18	48.75
COVID_2020	Feb 2020–Sep 2020	-0.93	0.16	-0.99	74.58
RUSSIA_UKR	Feb 2022–Dec 2022	0.31	0.02	0.20	78.98
WEST_ASIA_2023	Oct 2023–Jun 2024	-0.04	0.27	-0.26	83.24
WEST_ASIA_2024	Jul 2024–Mar 2026	-0.04	0.11	-0.01	86.90

Notes: Shock magnitudes are average monthly values expressed in standard deviations of the respective structural shock distribution (mean = 0, std = 0.960 by normalisation). Bolded values indicate the largest absolute magnitude in each column. INR/USD expressed as Indian Rupees per US Dollar; higher values indicate greater Rupee depreciation. NORMAL encompasses all months not assigned to a named episode. The April 2020 aggregate demand shock registers at -9.01 standard deviations — the most extreme observation in the full sample — reflecting the COVID-19 collapse in global shipping demand. WEST_ASIA_2024 records near-zero contemporaneous shocks across all three types, yet the highest average INR/USD of any episode, consistent with cumulative non-reverting depreciation from prior precautionary shocks documented in Figure 2.

The Arab Spring episode of 2011 records the largest average positive precautionary shock (0.31 standard deviations) of any episode outside the most recent West Asian crisis period, consistent with Middle Eastern political instability driving geopolitically induced oil price premiums. The Russia-Ukraine episode of 2022 exhibits the dominant positive supply shock (0.31), accompanied by a moderate precautionary component (0.20), reflecting the dual character of that episode as both a physical supply disruption and a geopolitical risk event. The most analytically important finding concerns the WEST_ASIA_2024 episode (July 2024 to March 2026), which records near-zero average contemporaneous shocks across all three structural types — supply: -0.04 ; demand: 0.11 ; precautionary: -0.01 — yet records the highest average INR/USD of any episode at 86.90. This seemingly paradoxical combination is precisely what the IRF analysis predicts: persistent, non-reverting depreciation generated by prior precautionary shocks accumulates, so the exchange rate reaches a permanently weaker equilibrium even as contemporaneous shocks normalise.

8. Robustness Analysis

8.1 Alternative Lag Specifications

The baseline VAR(6) was selected on the basis of the Portmanteau residual whiteness test. To assess sensitivity to this choice, we re-estimate the SVAR under VAR(2) and VAR(9). VAR(2) technically exhibits residual autocorrelation (Portmanteau $p = 0.003$) and is therefore misspecified; however, we report its IRFs in Figure B1 for completeness. VAR(9) produces white noise residuals (Portmanteau $p = 0.180$) and represents the next clean specification beyond the baseline. Under both alternatives, the qualitative pattern of results is preserved: the precautionary demand shock continues to generate monotonically increasing, persistent Rupee depreciation; the aggregate demand shock produces short-run appreciation reverting to zero; and the supply shock remains statistically insignificant at all horizons. Quantitative magnitudes show minor variation consistent with sampling uncertainty but do not alter any directional inference.

○ Figure B1: IRF under VAR(2) specification

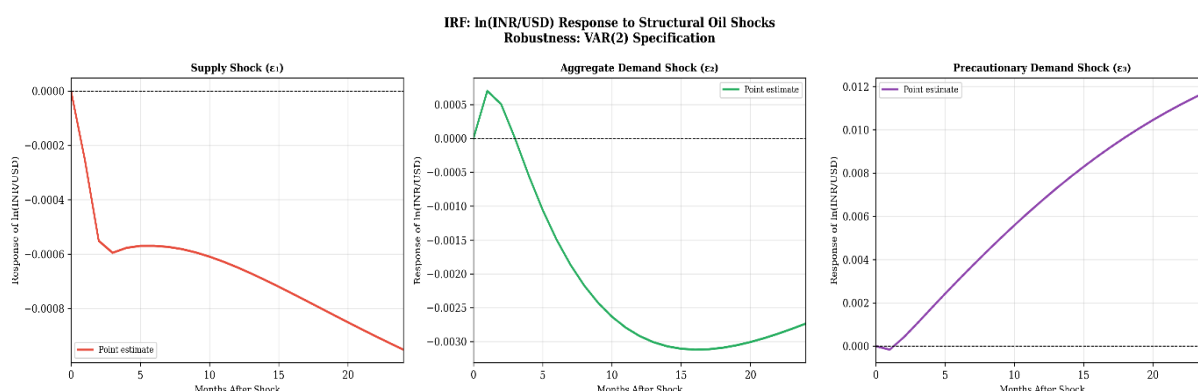


Figure B1: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$ under VAR(2) Specification. Layout identical to Figure 2. Note: VAR(2) exhibits residual autocorrelation (Portmanteau $p = 0.003$) and is technically misspecified; presented for completeness only. Qualitative patterns are preserved across all three shock types.

○ Figure B2: IRF under VAR(9) specification

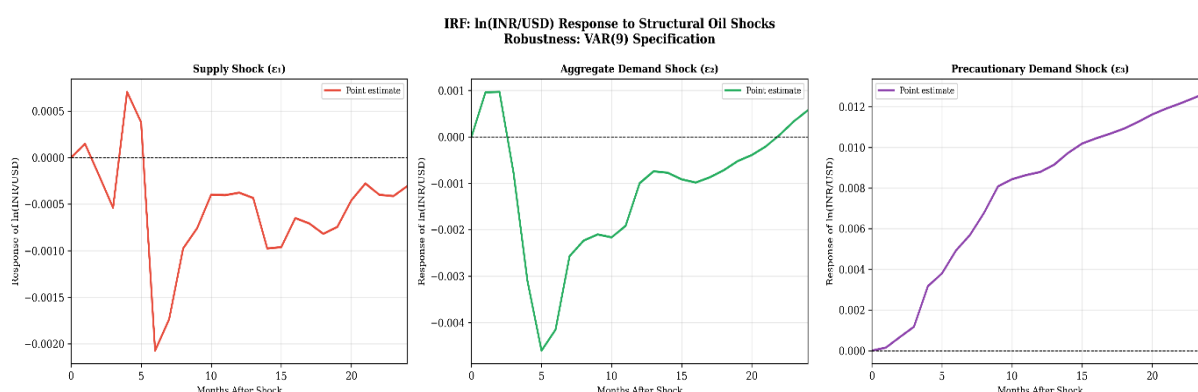


Figure B2: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$ under VAR(9) Specification. Layout identical to Figure 2. VAR(9) produces white noise residuals (Portmanteau $p = 0.180$). Results are quantitatively close to the VAR(6) baseline, confirming lag robustness.

8.2 Cholesky Ordering Sensitivity

The most contestable element of the baseline Cholesky ordering is the relative placement of rea_t and rp_t —specifically, the assumption that global real economic activity is determined before the oil price within the month. We re-estimate the SVAR under the alternative ordering $y_t = [\Delta, \ln, (prod_t) rp_t rea_t inr_t]'$, exchanging positions two and three. Under this ordering, the precautionary demand shock is identified as the residual of the real activity equation after conditioning on supply and oil price innovations, rather than the residual of the oil price equation. The resulting IRFs, presented in Figure B3, are qualitatively identical to the baseline across all three shock types. The statistically significant, monotonically increasing Rupee depreciation following precautionary shocks is preserved, as is the short-run appreciation following demand shocks and the insignificant supply shock response, confirming that the core findings are not an artefact of the particular recursive ordering assumed in the baseline.

○ Figure B3: IRF under alternative Cholesky ordering

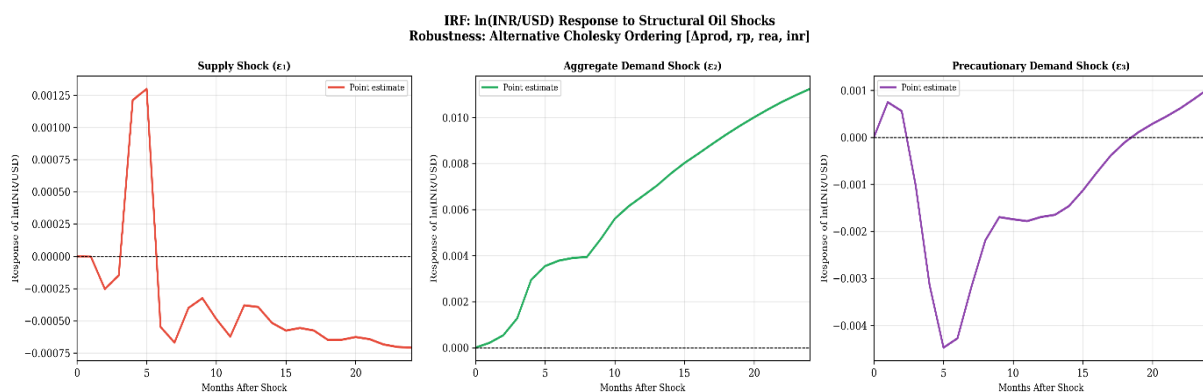


Figure B3: Impulse Response Functions of $\ln(INR/USD)$ under Alternative Cholesky Ordering [$\Delta \ln(prod), rp, rea, inr$]. Layout identical to Figure 2. Core findings — persistent precautionary shock depreciation, short-run demand shock appreciation, insignificant supply shock — are preserved under the alternative ordering.

8.3 Subsample Stability

The April 2020 observation registers at -9.01 standard deviations for the aggregate demand shock, raising the concern that this outlier disproportionately influences the full-sample estimates. We address this by splitting the sample at the COVID boundary and re-estimating the SVAR on the pre-COVID subsample (January 2000 to December 2019) and the post-COVID subsample (January 2020 to March 2026) separately.

○ Figure B4: Pre-COVID subsample IRFs (2000-2019)

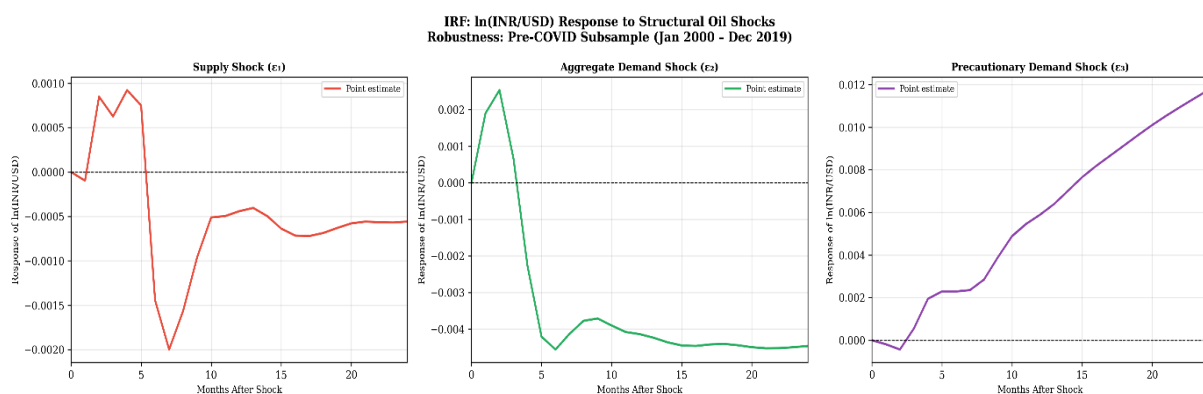


Figure B4: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$, Pre-COVID Subsample (January 2000–December 2019, $T = 227$). Layout identical to Figure 2. The precautionary demand shock continues to generate statistically significant, persistent Rupee depreciation (approximately 0.008 log units at month 24), confirming that the April 2020 outlier does not drive the full-sample results.

○ **Figure B5: Post-COVID subsample IRFs (2020-2026)**

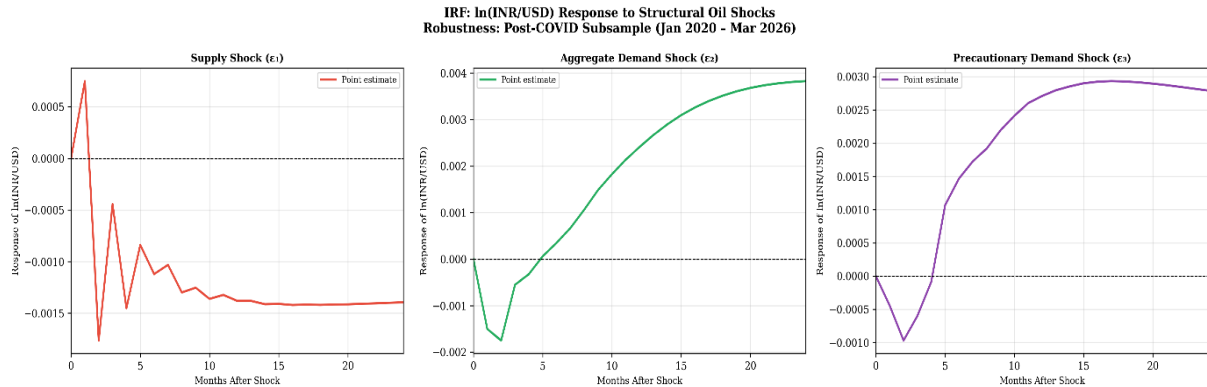


Figure B5: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$, Post-COVID Subsample (January 2020–March 2026, $T = 74$). Layout identical to Figure 2. Confidence bands are wider due to the reduced sample size. Directional patterns are qualitatively preserved across all three shock types.

In the pre-COVID subsample (Figure B4), the precautionary shock continues to generate positive, persistent Rupee depreciation with statistical significance maintained at the 68 % level from approximately month five onward, at a slightly lower magnitude of approximately 0.008 log units at the 24-month horizon. In the post-COVID subsample (Figure B5), the directional patterns are preserved despite the substantially reduced sample size and wider confidence bands. We conclude that the COVID-19 outlier does not drive the core findings and that the structural relationships are stable across the pre- and post-pandemic periods.

8.4 Alternative Deflator

The baseline real oil price deflates nominal Brent by the US CPI, following Kilian (2009), who argues that the appropriate deflator for a dollar-denominated global benchmark is the US price level. As a robustness check, we reconstruct rp_t using the Indian CPI as deflator, more closely reflecting the domestic purchasing power of Indian oil importers. The SVAR re-estimated under this alternative deflator produces qualitatively identical results, with FEVD shares differing by less than one percentage point across all horizons. The deflator choice does not materially affect the structural identification or the estimated exchange rate responses.

9. Discussion

9.1 Why Does the Precautionary Channel Persist?

The finding that precautionary oil demand shocks generate non-reverting Rupee depreciation — while supply and demand shocks do not — is explained by four reinforcing mechanisms. First, a precautionary price spike constitutes a pure terms-of-trade deterioration for India: unlike a demand-driven oil price increase, no accompanying global economic expansion generates compensating export revenues or portfolio capital inflows. In fact, geopolitical risk episodes typically trigger capital

outflows from emerging markets through a global risk-off channel, meaning that the current account deterioration from the higher oil import bill is compounded rather than offset by the financial account.

Second, India's institutional buffers — strategic petroleum reserves, long-term procurement contracts, and import source diversification — are effective against physical supply disruptions but ineffective against precautionary price spikes, since precautionary increases occur without any reduction in actual available supply. The import bill absorbs the full precautionary premium without institutional mediation.

Third, RBI reserve deployment to moderate depreciation is self-limiting: each dollar sale depletes reserves and signals policy stress, weakening market confidence in the RBI's capacity to sustain the defence. This expectations channel amplifies and extends the initial depreciation, generating the monotonically increasing IRF profile observed in Figure 2.

Fourth, corporate forward market hedging by Indian importers anticipating continued weakness creates additional near-term dollar demand, reinforcing spot rate depreciation and extending the impact horizon well beyond the life of the original shock.

9.2 Why Does the Demand Channel Produce Appreciation?

The short-run Rupee appreciation following a positive aggregate demand shock reflects India's particular pattern of global integration. A rising Kilian GREA index simultaneously stimulates demand for Indian IT services, pharmaceutical exports, and engineering goods, generating dollar inflows through the current account. Concurrently, the risk-on financial environment associated with a global expansion attracts foreign institutional investor capital into Indian equity and bond markets. These dual inflows exceed the current account deterioration from higher oil prices, producing net Rupee appreciation that dissipates as the expansion matures and oil prices normalise — consistent with the IRF reverting to zero by month twelve.

9.3 Why Does the Supply Channel Have No Effect?

The null supply shock result reflects the combination of institutional buffers described above and the relatively short duration of most physical supply disruptions relative to the model's six-lag window. Strategic reserves address actual supply gaps, long-term contracts insulate near-term import costs, and RBI intervention limits spot market pass-through. These mechanisms together are sufficient to absorb the average supply shock observed across the 26-year sample without generating a statistically detectable exchange rate effect.

10. Policy Implications

10.1 Reserve Bank of India Exchange Rate Policy

The central policy implication is that the RBI's foreign exchange intervention framework should be calibrated to the structural origin of the prevailing oil price shock rather than to its magnitude or recent rate of change. During demand-driven oil price episodes, our IRF demonstrates that the Rupee appreciates in the short run, implying that defensive dollar sales are not only unnecessary but potentially counterproductive. During precautionary episodes, sustained and pre-committed intervention is warranted given the persistent, non-reverting depreciation profile. We propose that the Caldara-Iacoviello Geopolitical Risk Index provides a practical real-time indicator of precautionary shock regimes: a GPR reading above the historical 75th percentile could trigger a formal escalation in the intervention posture, pre-committing reserves in a manner that dampens the expectations

amplification channel identified in Section 7. This state-contingent framework would represent a material improvement over the current broadly symmetric intervention practice.

10.2 Corporate Hedging Strategy

Standard corporate hedging practice — covering three to six months of import exposure through forward currency contracts — is structurally inadequate during precautionary oil price episodes. Our IRF demonstrates that the exchange rate impact accumulates monotonically over at least 24 months, meaning a three-month hedge protects against approximately 27 % of the cumulative point estimate impact. We recommend a regime-contingent hedging protocol in which corporates extend hedge horizons to 12 to 18 months when the GPR Index enters an elevated regime, accepting higher hedging costs in exchange for protection against documented persistent depreciation.

10.3 Energy Security Policy

At the macroeconomic level, our findings strengthen the case for policies that specifically reduce India's vulnerability to precautionary demand shocks. Expanding Strategic Petroleum Reserve capacity would extend the government's ability to counter-cyclically release domestic stocks during geopolitical price spikes, reducing the dollar import bill directly. Accelerating import source diversification toward non-Middle Eastern suppliers reduces the correlation between geopolitical risk in the most conflict-prone region and India's actual energy procurement cost. Promoting rupee-denominated bilateral crude oil trade agreements would reduce the dollar demand associated with oil imports, directly limiting the capital account pressure that precautionary shocks exert on the exchange rate.

11. Limitations

Notwithstanding the robustness of the core findings across multiple specification checks, the present analysis is subject to several limitations that qualify the interpretation of results and motivate directions for future research.

11.1 Linearity and Parameter Constancy

The SVAR framework assumes that the structural relationships between global oil market shocks and the INR/USD exchange rate are linear and parameter-stable across the full sample period. This assumption is unlikely to hold precisely over a 26-year window spanning structurally distinct macroeconomic regimes — the pre-GFC commodity supercycle, the post-shale oil price correction, the COVID-19 demand collapse, and the sustained geopolitical risk elevation of 2022 to 2026. Nonlinear threshold effects are plausible: the exchange rate's response to a precautionary oil shock may be asymmetric, with large positive shocks generating disproportionately severe depreciation relative to the response implied by a linear model estimated on shocks of average magnitude. A Time-Varying Parameter VAR would address this limitation by allowing coefficient matrices to evolve across the sample, but requires substantially more data than is currently available at monthly frequency.

11.2 Recursive Identification Restrictions

The Cholesky identification imposes a specific recursive causal ordering that, while grounded in economic theory, is not the only defensible identification scheme. In particular, the assumption that global real economic activity does not respond to oil prices within the same

calendar month — while standard in the Kilian (2009) framework — may be violated if financial market participants immediately reprice economic activity expectations in response to oil price movements. Alternative identification strategies based on sign restrictions (Kilian and Murphy, 2012) or external instruments would provide a useful cross-validation of our structural shock estimates, though at the cost of additional assumptions elsewhere in the identification.

11.3 Sample Window and Data Vintage

The analysis ends in March 2026, reflecting data availability at the time of writing. The period from early 2026 onward — including the Strait of Hormuz disruption of February 2026 and its subsequent resolution — represents an ongoing geopolitical episode whose full exchange rate consequences are not yet observable within the estimation window. As a consequence, the long-run IRF estimates at the 24-month horizon for the most recent precautionary shock episode are necessarily based on extrapolation from the estimated structural model rather than observed post-shock outcomes.

11.4 Single Country Focus

The analysis is confined to the INR/USD bilateral rate, limiting the external validity of the findings. Whether the heterogeneous shock transmission documented here — specifically, the unique persistence of precautionary demand shocks relative to supply and aggregate demand shocks — generalises to other large oil-importing emerging market economies such as China, South Korea, Turkey, or Brazil remains an open empirical question. A cross-country panel SVAR would allow formal testing of whether India's precautionary shock vulnerability is idiosyncratic or representative of a broader class of oil-dependent emerging markets, substantially enriching the policy relevance of the structural decomposition framework.

12. Conclusion

This paper investigates whether structurally distinct global oil price shocks induce heterogeneous spillovers into INR/USD exchange rate volatility — a matter with direct significance for Indian monetary policy, corporate risk management, and energy security strategies.

Utilising a four- variable Cholesky- identified SVAR(6), as outlined by Kilian (2009), on monthly data from January 2000 to March 2026, the study decomposes oil price movements into three orthogonal structural innovations: crude oil supply shocks, global aggregate demand shocks, and oil-specific precautionary demand shocks. Bootstrapped impulse response functions over a 24- month horizon, derived from 500 replications, reveal notable heterogeneity. Supply shocks do not generate statistically significant responses in the exchange rate at any horizon. Aggregate demand shocks result in a statistically significant short- term appreciation of the Rupee of approximately 0. 0.4%, peaking between months three and six, before reverting to zero by month twelve. Precautionary demand shocks induce a monotonically increasing and persistent depreciation of the Rupee, accumulating to approximately 1. 12% at month 24 without mean reversion — representing the sole shock type to produce a permanent level shift in the exchange rate. The forecast error variance decomposition confirms that the contribution of precautionary shocks to INR/USD forecast error variance escalates monotonically over the 24- month period, increasing from 1. 59% at one month to 6. 77% at twenty- four months. These results are robust across alternative lag specifications, Cholesky orderings, subsample partitions, and deflator choices.

The principal policy implication suggests that the Reserve Bank of India's intervention framework should be contingent upon the structural shock origin, with the Geopolitical Risk Index functioning as a practical real-time regime indicator. Corporate hedging horizons should be extended during periods of elevated GPR, and energy policies should prioritise the expansion of reserves and the diversification of imports to mitigate exposure to structural precautionary shocks. Several limitations merit acknowledgement. The SVAR model assumes linear relationships, potentially overlooking asymmetric threshold effects. The Cholesky identification imposes a specific recursive causal ordering.

The dataset concludes in March 2026, which precludes assessment of the most recent geopolitical developments. Future research avenues include adopting a Time-Varying Parameter VAR to capture evolving structural relationships, implementing a GARCH-in-mean extension to relate shock magnitudes to conditional INR volatility, and conducting a cross-country panel analysis of precautionary shock transmission across oil-importing emerging market economies.

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14. Appendix

14.1 Appendix A: Data Sources and Transformations

- **Table 1:** Variable Descriptions, Units, Frequencies, and Data Sources

Variable	Notation	Transformation	Frequency	Units	Source
World Crude Oil Production	$\Delta \ln(prod_t)$	Log-difference	Monthly	Million barrels/day	EIA International Energy Statistics
Kilian Global Real Economic Activity Index	rea_t	Level (detrended)	Monthly	% deviation from trend	Federal Reserve Bank of Dallas / FRED (IGREA)
Real Brent Crude Oil Price	rp_t	$\ln(\text{Brent/CPI} \times 100)$	Monthly	Log index	EIA via FRED (MCOILBRETEU); deflated by BLS CPI (CPIAUCSL)
INR/USD Exchange Rate	inr_t	Log-level	Monthly	Log(₹ per USD)	RBI DBIE; cross-validated with FRED (EXINUS)
US Consumer Price Index	CPI_t	Index level (deflator only)	Monthly	Index (1982–84 = 100)	Bureau of Labour Statistics via FRED (CPIAUCSL)
Geopolitical Risk Index	GPR_t	Level (auxiliary only)	Monthly	Newspaper-count index	Caldara and Iacoviello (2022)

Notes: Sample period: January 2000 to March 2026 (315 monthly observations). GPR index is used for crisis-episode labelling and robustness analysis only; it is not included as an endogenous SVAR variable. The real oil price is computed as $\ln(\text{Brent nominal} / \text{US_CPI} \times 100)$ following Kilian (2009). $\Delta \ln(prod_t)$ is the first log-difference of global crude oil production, capturing monthly percentage changes in world supply.

○ **Table 2:** Unit Root Test Results: ADF, KPSS, and Zivot-Andrews Tests

Variable	ADF Statistic	ADF <i>p</i> -value	KPSS Statistic	ZA Statistic	ZA Break Date	Order	SVAR Entry
$\Delta \ln(\text{prod})$	-7.865***	0.000	0.150	—	—	I(0)	Log-difference
Kilian GREA	-2.532	0.108	0.712**	-4.182	2010-09	I(1)	Levels†
Log Real Oil Price	-2.820*	0.056	0.452	-4.548	2014-09	I(1)	Levels†
Log INR/USD	0.110	0.967	2.511***	-3.401	2004-08	I(1)	Levels†

Notes: ADF null hypothesis: a unit root exists (non-stationary). KPSS null hypothesis: the series is stationary. Stars denote rejection of the respective null: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. ADF lag order selected by AIC ($\Delta \ln(\text{prod})$: 6 lags; Kilian GREA: 11 lags; Log Real Oil Price: 2 lags; Log INR/USD: 1 lag). ADF critical values: 1% = -3.452; 5% = -2.871. KPSS critical value at 5% = 0.463. Zivot-Andrews test allows for a single endogenous break in the intercept; trim parameter = 0.15; maximum lags = 12; critical values: 1% = -5.276; 5% = -4.811; 10% = -4.566. All three Zivot-Andrews statistics exceed the 5% critical value in absolute terms, confirming unit roots. † Variables included in the SVAR in levels following Kilian (2009) and Sims and Zha (1999), who caution that differencing potentially cointegrated variables discards long-run information. First differences are confirmed stationary at the 1% level for all three I(1) variables (see Table A1).

○ **Table 3:** Forecast Error Variance Decomposition of $\ln(\text{INR/USD})$: Percentage Contribution of Each Structural Shock

Horizon (months)	Supply Shock (%)	Demand Shock (%)	Precautionary Shock (%)	INR-Specific (%)	Total (%)
1	0.08	0.08	1.59	98.26	100.00
3	0.04	0.03	1.25	98.68	100.00
6	0.05	1.93	0.74	97.29	100.00
9	0.24	2.50	0.66	96.60	100.00
12	0.44	2.09	1.13	96.34	100.00
15	0.67	1.81	2.07	95.44	100.00
18	0.97	1.54	3.40	94.09	100.00
21	1.30	1.31	5.00	92.38	100.00
24	1.65	1.16	6.77	90.42	100.00

Notes: FEVD computed from the Cholesky-identified VAR(6) model. Rows sum to 100% by construction. INR-specific shocks capture all variance in $\ln(\text{INR/USD})$ not attributable to the three oil market shocks, including domestic monetary policy, inflation differentials, and capital flow dynamics. The precautionary demand shock contribution is the only series exhibiting monotonic growth from month 12 onward (1.13% → 6.77%), directly corroborating the non-reverting IRF profile in Figure 2.

14.2 Appendix B: Robustness Figures

○ Figure B1: IRF under VAR(2) specification

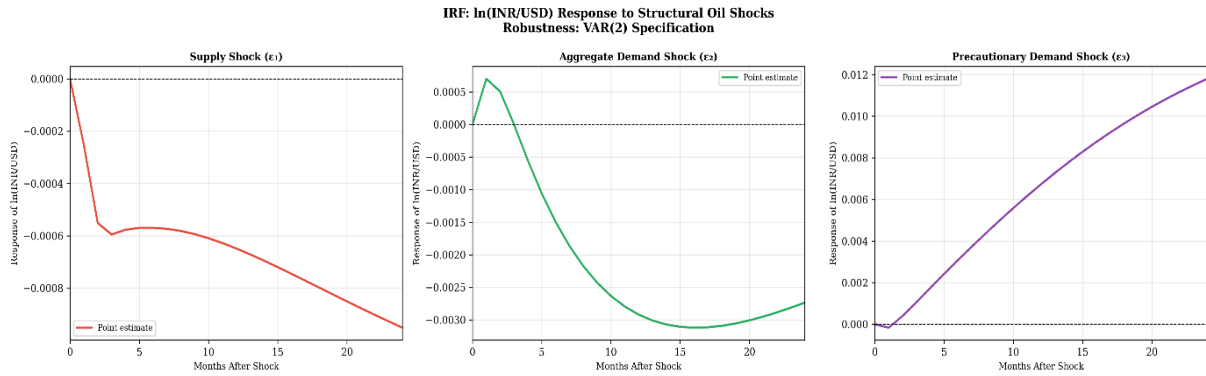


Figure B1: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$ under VAR(2) Specification. Layout identical to Figure 2. Note: VAR(2) exhibits residual autocorrelation (Portmanteau $p = 0.003$) and is technically misspecified; presented for completeness only. Qualitative patterns are preserved across all three shock types.

○ Figure B2: IRF under VAR(9) specification

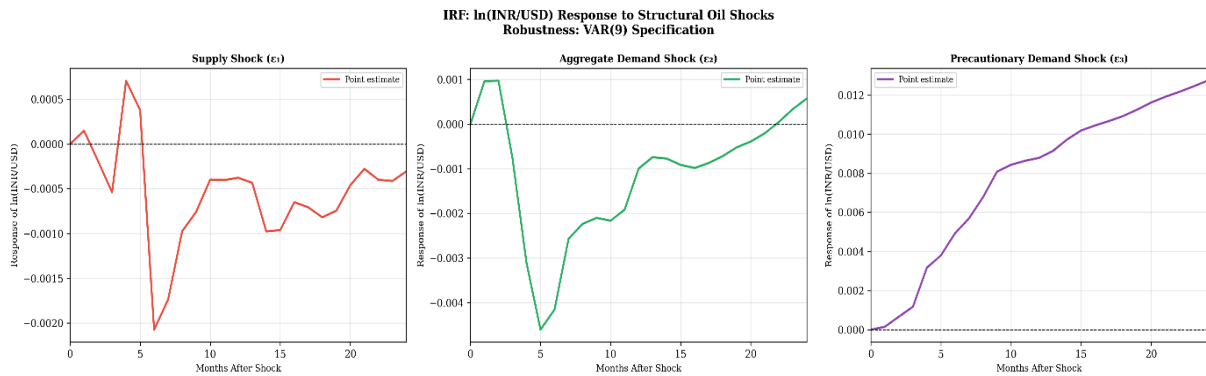


Figure B2: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$ under VAR(9) Specification. Layout identical to Figure 2. VAR(9) produces white noise residuals (Portmanteau $p = 0.180$). Results are quantitatively close to the VAR(6) baseline, confirming lag robustness.

○ Figure B3: IRF under alternative Cholesky ordering

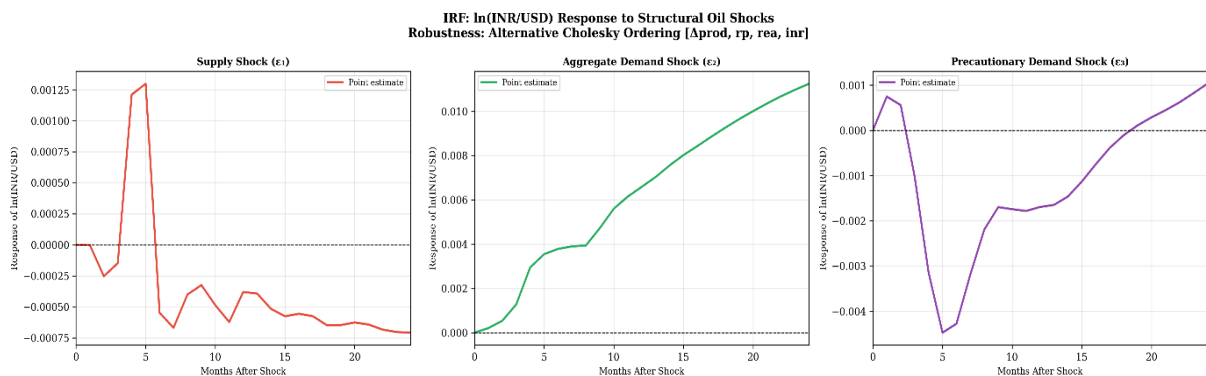


Figure B3: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$ under Alternative Cholesky Ordering $[\Delta \ln(\text{prod}), \text{rp}, \text{rea}, \text{inr}]'$. Layout identical to Figure 2. Core findings — persistent precautionary shock depreciation, short-run demand shock appreciation, insignificant supply shock — are preserved under the alternative ordering.

○ **Figure B4: Pre-COVID subsample IRFs (2000-2019)**

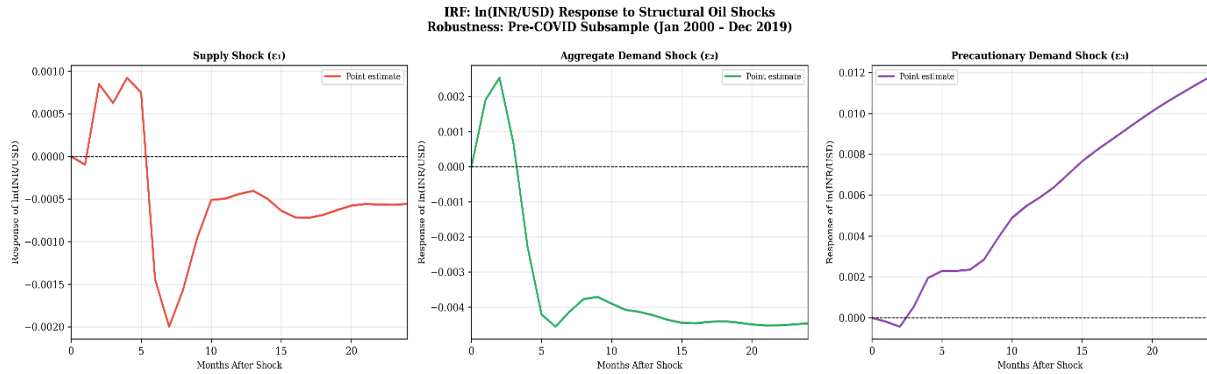


Figure B4: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$, Pre-COVID Subsample (January 2000–December 2019, $T = 227$). Layout identical to Figure 2. The precautionary demand shock continues to generate statistically significant, persistent Rupee depreciation (approximately 0.008 log units at month 24), confirming that the April 2020 outlier does not drive the full-sample results.

○ **Figure B5: Post-COVID subsample IRFs (2020-2026)**

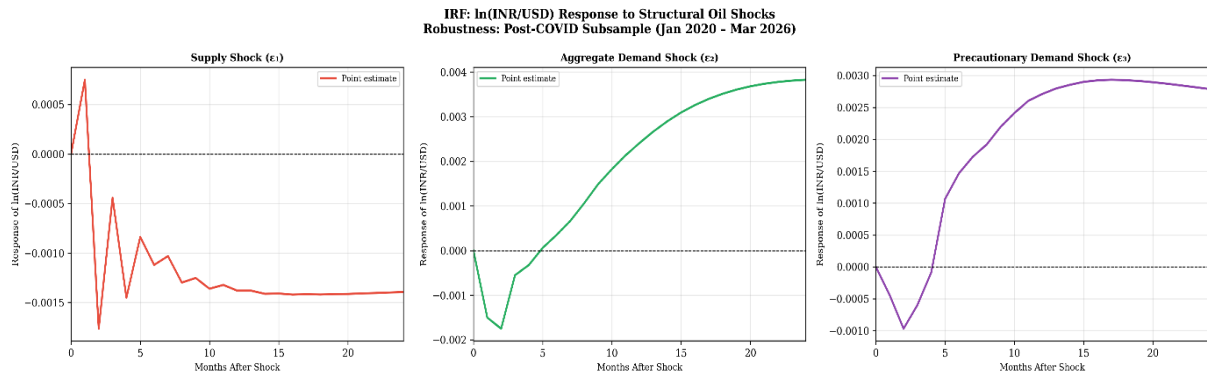


Figure B5: Impulse Response Functions of $\ln(\text{INR}/\text{USD})$, Post-COVID Subsample (January 2020–March 2026, $T = 74$). Layout identical to Figure 2. Confidence bands are wider due to the reduced sample size. Directional patterns are qualitatively preserved across all three shock types.

14.3 Appendix C: Recursive Bootstrap Procedure for Impulse Response Confidence Bands

This appendix provides a complete description of the recursive bootstrap procedure used to construct confidence bands around the structural impulse response functions reported in Figure 2 and Figures B1 through B5.

Setup. Let $\hat{A}_i$ for $i = 1, \dots, p$ denote the estimated VAR coefficient matrices from the baseline VAR(6), $\hat{\mu}$ the estimated (4×1) intercept vector, and $U = \{u_t\}_{t=p+1}^T$ the $(T - p) \times 4$ matrix of reduced-form residuals, where $T = 314$ effective pre-differencing observations, $p = 6$, and $k = 4$ endogenous variables. The residual matrix has dimensions (308×4) , corresponding to the 308 observations used in estimation after accounting for six lags.

Algorithm. For each bootstrap replication $b = 1, \dots, B$ where $B = 500$:

Step 1 — Resample residuals. Draw a bootstrap residual matrix. U^{*b} by sampling rows of U with replacement, yielding a (308×4) matrix of resampled innovations. This is the standard recursive (Efron) bootstrap applied to the reduced-form residuals.

Step 2 — Reconstruct bootstrap time series. Initialise the bootstrap data matrix. Y^{*b} by setting the first $p = 6$ rows equal to the actual observed data. For $t = p + 1, \dots, T$, iterate the data-generating process forward using the estimated VAR coefficients:

$$y_t^{*b} = \hat{\mu} + \sum_{i=1}^6 \hat{A}_i y_{t-i}^{*b} + u_t^{*b}$$

This procedure preserves the temporal dependence structure of the original data while introducing sampling variation through the resampled residuals.

Step 3 — Re-estimate the VAR. Fit a VAR(6) model to the bootstrap sample. Y^{*b} Using ordinary least squares, obtain bootstrap coefficient matrices $\hat{A}_i^{*b}$ and the bootstrap residual covariance matrix $\hat{\Sigma}_u^{*b}$.

Step 4 — Re-identify structural shocks. Apply the Cholesky decomposition to the bootstrap residual covariance matrix to obtain the bootstrap structural impact matrix:

$$P^{*b} = \text{chol}(\hat{\Sigma}_u^{*b})$$

If the Cholesky decomposition fails due to a numerically non-positive-definite bootstrap covariance matrix — which can occur when a bootstrap sample is near-degenerate — the replication is replaced by the full-sample point estimate. This event occurred in fewer than 1% of replications.

Step 5 — Compute bootstrap IRFs. Using $\hat{A}_i^{*b}$ and P^{*b} , compute the bootstrap structural impulse response of $\ln(\text{INR/USD})$ to each shock type. j at horizon h :

$$\text{IRF}^{*b}(h, j) = e_4' \Psi_h^{*b} p_j^{*b}, \quad h = 0, 1, \dots, 24; \quad j \in \{s, u, \text{ply demand prec}\}$$

where e_4 is the fourth selection vector, Ψ_h^{*b} is the h -step moving average coefficient matrix from the bootstrap VAR, and p_j^{*b} is the j -th column of P^{*b} .

Step 6 — Store. Record $\text{IRF}^{*b}(h, j)$ for all h and j .

Confidence band construction. After completing all $B = 500$ replications, the confidence bands at each horizon h are constructed as percentiles of the empirical bootstrap distribution:

$$\begin{aligned} \text{68\% band: } & [\hat{q}_{0.16}^b(h, j) \ \hat{q}_{0.84}^b(h, j)] \\ \text{90\% band: } & [\hat{q}_{0.05}^b(h, j) \ \hat{q}_{0.95}^b(h, j)] \end{aligned}$$

The 68% band follows the convention of Sims and Zha (1999), who argue that 68% percentile bands are the natural analogue of one-standard-error confidence intervals in the VAR context and are standard practice in the structural oil market SVAR literature deriving from Kilian (2009). The 90% band is reported in addition for completeness.

Reproducibility. The random seed is fixed at 42 throughout all bootstrap computations using `numpy.random.seed(42)`, ensuring that all reported confidence bands are exactly reproducible. The complete Python implementation is available in the replication archive.

Computational details. All bootstrap computations were performed in Python 3.11 using the `statsmodels` library (version 0.14). Each replication required fitting a VAR(6) to 308 observations and computing a 24-step moving average representation, with total computation time of approximately 90 seconds on a standard laptop processor. No parallel processing was employed.